QOC Design Space Analysis

Oct 26, 2026 · @Jishnu

Where this sits in the process

This QOC compares LectureFlow, Student Saviour and SpokeSafe against one fixed set of criteria: the criteria bank below. Every mark in every matrix cites a bank criterion by its ID and gives a reason from that concept's own document. There are no weights, scores or decisions yet.

QOC is mostly a Define-to-Ideate tool. Q1 also shapes Layer 1, because it decides who we research. Every matrix will be revisited once questionnaire and interview results are in.

Marking rule, applied the same way to every concept:

- **+** = the concept's own document describes a concrete feature or mechanism that meets the criterion.
- **−** = the concept works against the criterion, and its document offers no fix.
- **?** = not yet addressed, dependent on research, or a named risk whose proposed fix is untested.
- Marks are never totalled. They show trade-offs to discuss, not a ranking.

Bias review

Earlier versions of this doc, drafted with an AI assistant over two days of chat, favoured SpokeSafe. The fixes below remove the chat's influence by making module material the single source of truth for criteria.

Bias found	Evidence from the chat	Fix in this version
Anchoring on early ideas	Cycling and safety ideas were suggested first and called the "strongest" before any research. The assistant wrote SpokeSafe's document; the group wrote LectureFlow's.	Each concept is judged only on its own document, against fixed criteria.

Bias found	Evidence from the chat	Fix in this version
Criteria chosen to suit one option	The first Q1 criteria (rich sensing, novelty) were picked in chat and suited SpokeSafe. Criteria where LectureFlow is strong were missing.	The criteria bank comes from the brief and lectures, and was fixed before marking. Q1 uses module criteria only.
Inconsistent marking	LectureFlow got − for sensing, though the brief accepts data from APIs; − for prototyping where SpokeSafe got ?. SpokeSafe got + for privacy and novelty without evidence.	One written marking rule, with a cited reason in every cell.
Uneven description	LectureFlow's "against" points were longer and more detailed than the others'.	Every cell gets one short reason, in the same format for all three concepts.
Outside theory presented as course content	Fogg's behaviour model, self-determination theory and adaptive interventions came from the chat, not the module notes.	Moved to a labelled supplementary list, not used in any matrix.
Extra attention to the safety concepts	A whole question (Q1a) scoped the two safety ideas. Questionnaire v1 has 5 safety questions (Q7–Q11) and none on lectures.	Q1a is parked. The research section flags the questionnaire gap.

Remaining risk: bias the other way. Most of the group now favours LectureFlow, and the same marking rule must stop us marking it up. The personas lecture's warning against "self-referential design" applies to the group as much as to the assistant: we are not the user.

Criteria bank

Every criterion used in a matrix comes from this bank. It has 23 module criteria taken from the assignment brief and the module lectures, plus 3 that our own questionnaire will answer. Following the QOC lecture, each is phrased as a positive.

Module criteria

ID	Criterion	Source
B1	Acts on its own initiative, not only when the app is opened	Brief
B2	Integrates data from at least one source beyond user input: sensors or external APIs	Brief
B3	Intervenes with an action, not just information	Brief
B4	Gives users a better experience than they had before	Brief; Understanding Users ("what are people doing now, how can we improve on that?")
B5	Has significant impact for its target users	Brief
B6	Builds on prior experiences and respects social expectations	Brief (UX elements)
B7	Supports personal goals and progress, with joy, humour or gamification where it fits	Brief (UX elements)
U1	Is effective and useful: helps users reach their goal	Week 1 Intro: usability goals (effectiveness, utility)
U2	Is efficient, easy to learn and easy to remember	Week 1 Intro: efficiency, learnability, memorability
U3	Is safe: avoids errors and harm to users	Week 1 Intro: safety
U4	Is fun, engaging and enjoyable	Week 1 Intro: user experience goals
H1	Fits users' capabilities and individual differences	Human in HCI (individual differences in perception); Understanding Users (user characteristics)
H2	Keeps short-term memory load low	Human in HCI (7±2 chunks, chunking)
H3	Reduces what users must recall	Human in HCI (long-term memory: recall and recognition)
H4	Gives a sense of closure	Human in HCI (closure)
H5	Matches users' mental models	Human in HCI (mental models)
H6	Gets attention without overloading it	Human in HCI (selective attention)

ID	Criterion	Source
C1	Fits the context of use: physical, social, organisational, technical	Understanding Users; Gulliksen et al. (holistic design)
C2	Collects only the data it needs	Questionnaire guidance ("is it necessary data, and consider GDPR")
C3	Reflects every stakeholder's concerns, not just the main user's	QOC lecture (criteria reflect stakeholder concerns)
P1	Representative users can be involved early and throughout	Gulliksen et al. (active user involvement); Understanding Users
P2	Can be prototyped early and evaluated with users	Gulliksen et al. (prototyping, evolutionary development)
P3	Improves on similar existing products	Understanding Users (researching similar products)

Group-research criteria

ID	Criterion	Source
G1	Stays within the number of notifications students will accept	Questionnaire v1, Q14
G2	Uses only data students say they would allow	Questionnaire v1, Q13
G3	Is welcomed as an alert the student didn't ask for	Questionnaire v1, Q12

Supplementary ideas (from the chat, not in the module notes)

These are not used in any matrix. Adopt one only if the group agrees and can cite a published source.

- Prompts when motivation and ability are both high (Fogg's behaviour model).
- Supports autonomy and competence (self-determination theory).
- Adapts to the individual over time (just-in-time adaptive interventions).
- Interrupts at transitions, not in the middle of a task.
- Avoids guilt, shame or anxiety.
- Still works when data is missing; is light on battery.

Q1: Which concept should we take forward?

Marked on module criteria alone, no concept dominates. All three meet the brief's core requirements (B1–B3). They differ on only two criteria: engagement (U4), where Student Saviour describes no feature, and reaching users (P1), where LectureFlow depends on recruiting students with ADHD. The criteria that would decide the choice (B4, B5, P3) are ? for all three until research is done.

Options

- **LectureFlow:** recorded lectures in short sections, scheduled around free time, for students with ADHD.
- **Student Saviour:** a personal safety companion for journeys, campus security alerts and check-ins.
- **SpokeSafe:** bike theft prevention, intervening when a student parks and if the bike moves.

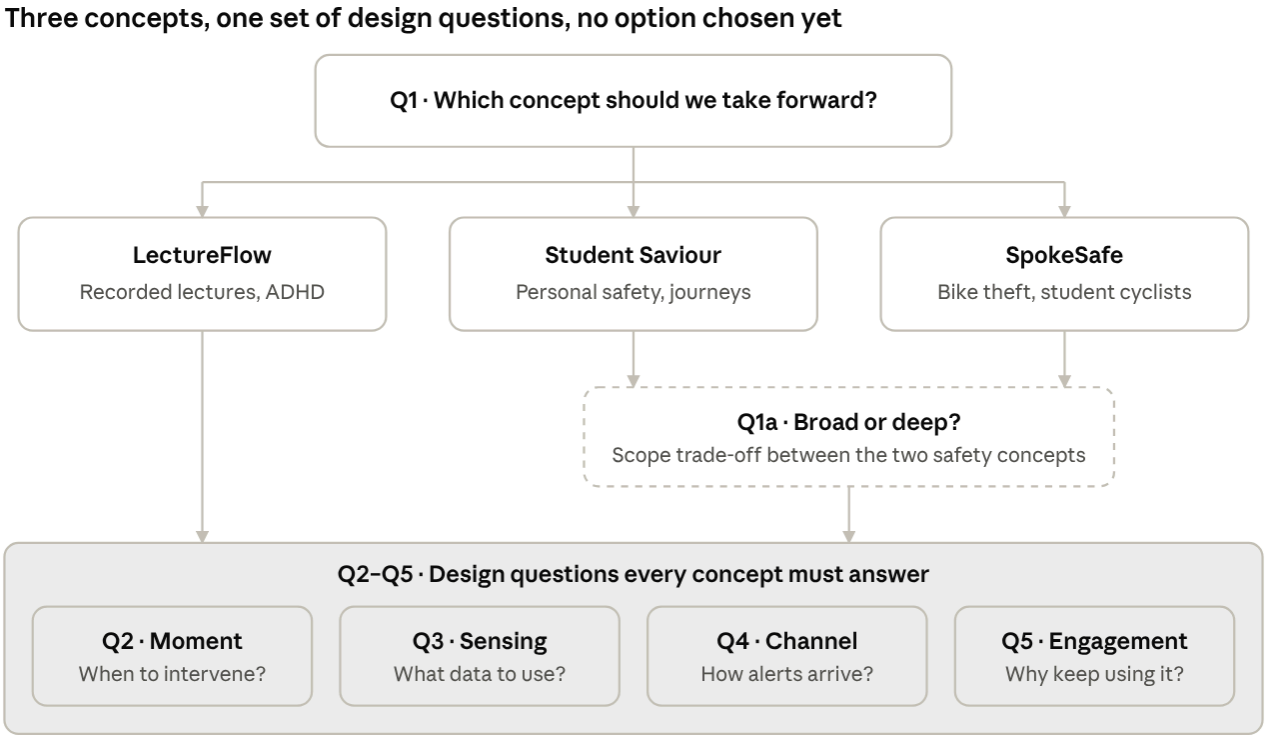
Section numbers (§) refer to each concept's own document.

Criterion	LectureFlow	Student Saviour	SpokeSafe
B1 Acts on its own initiative	+ §5: five trigger scenarios	+ §5: five trigger scenarios	+ §5: seven trigger scenarios
B2 Integrates data beyond user input	+ §4: timetable, recordings, calendar	+ §4: police.uk, transport API, GPS	+ §4: police.uk, OpenStreetMap, Bluetooth tag
B3 Intervenes with an action	+ §3.6: offers a specific section to watch now	+ §3.3–3.4: alternative routes, check-ins	+ §3.1–3.2: alternative rack, theft report
B4 Better experience than before	? §9: benefit to be shown by testing	? §9: benefit to be shown by testing	? §9: benefit to be shown by testing
B5 Significant impact for target users	? How common the problem is: not yet asked	? Questionnaire Q4, Q5, Q11	? Questionnaire Q7–Q9
U3 Safe: avoids errors and harm	? §10: AI summaries may be wrong; fix (original always available) untested	? §10.5: false sense of security; fix (no safety ratings) untested	? §10.4, 10.8: false alarms, confrontation; fixes untested

Criterion	LectureFlow	Student Saviour	SpokeSafe
U4 Fun, engaging, enjoyable	+ §3.3: knowledge checks, interactive summaries	? No engagement feature described	+ §3.6: impact feedback, club challenges
H1 Fits individual differences	+ §3.2, 3.4: speed, captions, audio or video	+ §3.2: customisable alert categories	+ §3.4: advice fitted to bike and lock
C2 Collects only the data it needs	? Not yet addressed; no diagnosis is required	? §10.3: location tracking; fixes proposed	? §10.6: location and ride detection; fixes proposed
P1 Representative users can be involved	? §7.1: depends on reaching students with ADHD	+ Any student can take part	+ Student cyclists, including national-level contacts
P2 Can be prototyped and evaluated	? §10: recordings and timetable simulated	? §10.7: security feed simulated	? §11: tag and ride detection simulated
P3 Improves on similar products	? Competitor analysis planned (§7.1), not done	? Competitor analysis planned (§7.1), not done	? Competitor analysis planned (§7.1), not done

Group status: most of the group favours LectureFlow. The matrix doesn't contradict that choice, but it doesn't confirm it either. B4, B5, P1 and P3 still need evidence, and LectureFlow's B5 can't be answered until the questionnaire asks about lectures.

Question map



QOC question map · 3 concepts, 1 scope question, 4 shared design questions

All three concepts answer the same design questions, Q2 to Q5, so they can be compared on equal terms. Q1a applies only to the two safety concepts.

Q1a: How broad should a safety concept be? (parked)

This question is parked. It only matters if the group chooses Student Saviour or SpokeSafe. Its earlier matrix used criteria invented in chat, so the matrix has been removed rather than kept as evidence.

If a safety concept is chosen, argue these options against the criteria bank:

- **Broad:** Student Saviour, with bike theft removed.
- **Deep:** SpokeSafe on its own.
- **Broad with a deep module:** Student Saviour, with SpokeSafe as its bike module.
- **Deep first, then extend:** SpokeSafe now, journey features later.

Q2: When should each concept intervene?

Each concept gets four candidate moments, taken from the scenarios in its own document. Arguments use these bank criteria: B1, B3, C1, C2, C3, H1–H3, H6, U2, U3, P2 and G1.

Concept	Candidate moment	For	Against
LectureFlow	A free slot appears in the calendar (§5, scenario 2)	B1, B3: uses time that already exists	C1: relies on calendars being complete (to check); G1: repeated offers could exceed the notification limit
LectureFlow	A related seminar is coming up (§5, scenarios 1 and 4)	B3, C1: a clear reason to act, tied to the timetable	U3: may add last-minute pressure (to test with users)
LectureFlow	The student reopens an unfinished lecture (§5, scenario 3)	H3: a recap reduces what must be recalled; H6: low interruption	B1: starts with the user, so it isn't proactive on its own
LectureFlow	A repeated preference is detected (§5, scenario 5)	U2, H1: adapts to the individual and saves changing settings	P2: needs weeks of use data to prototype and test
Student Saviour	Entering an area with a security notice (§5, scenario 1)	B1, C1: right place, right time	P2: no known security feed; U3: may cause anxiety (to test)
Student Saviour	Planned bus cancelled on the way home (§5, scenario 2)	B3, U3: practical help at a stressful moment	B1: only works on journeys started in the app
Student Saviour	Expected arrival passes without a check-in (§5, scenario 4)	U3: the most critical safety moment	U3, C3: false alarms worry contacts
Student Saviour	Planning a route with a reported lighting fault (§5, scenario 5)	B3: prevents rather than reacts	P2: needs enough moderated reports
SpokeSafe	A ride ends away from home (§5, scenario 1)	B3, C1: the moment a parking choice is made	P2: ride-end detection is untested
SpokeSafe	The parked bike moves while the owner is away (§5, scenario 2)	U3: the only chance to respond to a theft in progress	U3: false alarms and confrontation risk

Concept	Candidate moment	For	Against
SpokeSafe	A theft is reported near the parked bike (§5, scenario 3)	B1, C1: timely and local	P2: needs enough student reports
SpokeSafe	Owner is home in the evening, bike still out (§5, scenario 4)	B1, B3: targets overnight risk	C2: needs to know where both home and the bike are

Q3: What should each concept sense?

The options come from the data-sources table (§4) in each concept's document. Arguments use these bank criteria: B2, C1, C2, C3, U3, P2 and G2.

Concept	Sensing or data option	For	Against
LectureFlow	University timetable	B2: authoritative source of lectures and seminars	P2: access depends on the university; may be simulated
LectureFlow	Lecture recordings and transcripts	B2: needed for sections and summaries	C3: lecturers' consent needed; U3: AI summaries may be wrong
LectureFlow	Phone calendar	B2: already on the phone	C1: how complete students keep it is unknown (to check)
LectureFlow	Playback activity	B2: precise progress and pause data	P2: only exists inside our own app
LectureFlow	Accelerometer and GPS (optional)	B2: could support movement-friendly listening	C2, G2: benefit unproven; adds location data
Student Saviour	GPS and geofencing	B2: core to journeys and location alerts	C2, G2: continuous location (§10.3)
Student Saviour	Police.uk crime data	B2: free and official	U3: 1–2 months old, approximate locations (§10.1)

Concept	Sensing or data option	For	Against
Student Saviour	Transport for West Midlands API	B2: live delays and cancellations	P2: only covers bus and train journeys
Student Saviour	University security notices	B2, C3: verified and campus-specific	P2: no known public feed; likely simulated (§10.7)
Student Saviour	Community reports	B2: current and precise	U3: need moderation; risk of misinformation (§10.6)
SpokeSafe	Phone activity recognition	B2: detects ride start and end with no extra hardware	P2: accuracy untested (§10.3)
SpokeSafe	Bluetooth motion tag	B2: the only way to detect the bike moving	P2: costs money; AirTags can't be read by other apps (§10.5)
SpokeSafe	Police.uk data plus student reports	B2: baseline from day one, kept current by reports	U3: same lag and moderation issues (§10.1, 10.9)
SpokeSafe	OpenStreetMap cycle parking	B2: lets the app suggest a better rack	P2: coverage of student areas to check
SpokeSafe	Sunset and weather data	B2: simple, reliable trigger	Secondary to the theft focus

Q4: How should interventions reach the user?

The same channel argues differently for each concept, because each user is in a different context of use (C1) when a prompt arrives: at a desk, walking home, or at a bike rack. Arguments use these bank criteria: B3, B6, C1, H1, H6, U3 and G1.

Channel	LectureFlow	Student Saviour	SpokeSafe
Phone notification	? B6: familiar; G1, H6: adds to notification volume, so its limit depends on Q14	+ C1: reaches the user mid-journey	? C1: the phone may be in a bag while locking up (to observe)

Channel	LectureFlow	Student Saviour	SpokeSafe
Smartwatch tap	? H6: a gentle nudge; H1: only for watch owners	? C1: discreet while walking; H1: only for watch owners	? C1: felt at the rack; H1: only for watch owners
Lock-screen card	+ H6: calmly shows lecture progress	+ H6: keeps journey status visible	− C1: only seen if the user checks the phone
Inside the app only	− B3: not proactive (useful for recaps)	− B3: not proactive	− B3: not proactive
Audible alarm	− C1: unsuitable in study settings	? U3: could escalate a missed check-in	? U3: justified for "bike moved", if false alarms are rare

We don't yet know how many students own a smartwatch. That decides H1 for the watch row in all three concepts.

Q5: How should each concept keep users engaged?

The options come from the brief's UX elements (B6, B7) and the Week 1 user experience goals (U4). Arguments use these bank criteria: B6, B7, U3, U4 and H4. Where a concept's document already describes the approach, the cell cites it.

Approach	LectureFlow	Student Saviour	SpokeSafe
Streaks	? B7: shows progress; U3: a broken streak may discourage (to test)	− B7: no clear daily goal to count	? B7: counts safe parks; U3: invites fake entries
Progress and personal goals	+ B7, H4: sections and lectures completed (§3.1)	? B7: little to progress towards	+ B7, H4: Bike Passport completeness (§3.4)
Peers and social features	? B6: studying alongside friends could help (to research)	+ B6: friends look out for each other (§3.4)	+ B6: club challenges (§3.6)
Community impact	? U4: learning is mostly individual	+ U4: verified reports help others (§3.6)	+ U4: "your reports warned riders" (§3.6)
Humour and delight	+ U4: a light, encouraging tone (§5, scenario 3)	? U3: risky next to a safety topic	? U4: could lighten lock-up tips

Approach	LectureFlow	Student Saviour	SpokeSafe
No game elements	? U4: some students may prefer a calm tool	? U3: safety may not suit games	? U4: may be forgotten once novelty fades

What research needs to settle

Each row below turns a ? in the matrices into a + or −. Once these are answered, we weight the criteria (High, Medium, Low), re-mark each matrix and choose.

Open question	Criteria	Concepts	Evidence
How common is each problem, and which do students most want help with?	B5	All	Questionnaire Q4, Q5
Do students struggle to engage with lectures or recordings?	B5	LectureFlow	Not covered: add a Q4 row, e.g. "Staying focused during lectures or recorded lectures"
How do students handle the problem now, and would the concept improve on it?	B4	All	Interviews; testing the prototype
How do existing products compare?	P3	All	Competitor analysis (each concept's §7.1)
Can we recruit representative users in time, ethically?	P1	LectureFlow	Disability services, neurodiversity societies; module ethics guidance
How common is bike theft among students who cycle?	B5	SpokeSafe, Student Saviour	Questionnaire Q7–Q9; police.uk data
How safe do students feel travelling at night?	B5	Student Saviour	Questionnaire Q11, compared with Q7
Do students welcome alerts they didn't ask for?	G3	All	Questionnaire Q12
Which data will students let an app use?	G2, C2	All	Questionnaire Q13

Open question	Criteria	Concepts	Evidence
How many notifications a day before they turn them off?	G1	All	Questionnaire Q14
Do students own a smartwatch?	H1	All	Not covered: add to Section D

The questionnaire is unbalanced: v1 has five safety questions (Q7–Q11) and none about lectures. Adding the lecture row to Q4 and the smartwatch question lets all three concepts be judged on the same evidence.